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REVIEW ARTICLE

Generative Adversarial Networks (GANs) for Medical Image Processing: Recent Advancements

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Abstract
Generative Adversarial Networks (GANs) constitute an advanced category of deep learning models that have significantly transformed the domain of generative modelling. They demonstrate a profound capability to produce realistic and high-quality synthetic data across various domains. Recently, GANs have also emerged as a powerful and innovative approach in medical image processing. Numerous scholarly investigations consistently highlight the superiority of GAN-based methodologies in this context. The generation of realistic synthetic images aims to advance segmentation precision, augment image quality, and facilitate multimodal analysis. These enhancements significantly bolster the analytical capabilities of medical professionals, leading to more precise diagnostic evaluations and the formulation of personalized treatment plans, thereby contributing to improved patient prognosis. In this work, we rigorously review the latest advancements in the application of Generative Adversarial Networks (GANs) within the domain of medical imaging, encompassing research published between 2018 and 2024. The corpus of literature selected for this review is derived from the most relevant and authoritative databases, including Elsevier, Springer, IEEE Xplore, and Google Scholar, among others. This review rigorously evaluates scholarly publications employing Generative Adversarial Networks (GANs) for the synthesis and generation of medical images, segmentation of medical imaging data, image-to-image translation in medical contexts, and denoising or reconstruction of medical imagery. The findings of this review present a thorough synthesis of contemporary applications of Generative Adversarial Networks (GANs) in the domain of medical imaging. This investigation serves as a prospective reference in the realm of GAN utilization for medical image processing, offering guidance and insights for current and future research endeavors.

1 Introduction

Medical imaging is indispensable in the diagnosis, treatment, and monitoring of diverse diseases and conditions. It facilitates automated interpretation and extraction of information from vast quantities of imaging data, thereby profoundly transforming the field of digital health care. Generative Adversarial Networks (GANs) are well-known deep learning technologies that have garnered considerable interest in contemporary research. GANs, which are a class of neural networks, have demonstrated exceptional success in producing realistic and excellent synthetic data. GANs offer distinctive skills and prospects within the domain of medical imaging. These methods possess the potential to mitigate various challenges and limitations associated with traditional techniques. A notable advantage lies in their ability to generate synthetic medical images, thereby complementing sparse training datasets and addressing data scarcity issues. GANs can enhance the generalization and

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robustness of deep learning models by synthesizing new datasets with good diversity and more training instances. GANs can also be applied for anomaly detection tasks by learning the typical distribution of medical images and looking for variations from this distribution. This capability is particularly advantageous for detecting anomalies, such as tumours or other pathological conditions, aiding physicians in formulating precise diagnoses and developing effective treatment strategies. GANs may also produce/generate realistic medical images that imitate certain illnesses, ailments, or imaging modalities. These generated images can be utilized for teaching, generating training data for uncommon or hard-to-find instances, or exploration of hypothetical scenarios aiding in the comprehension of disease progression and therapeutic responses. Additionally, GANs can help with image denoising, which improves the quality of medical images by reducing noise and enhancing visibility. This capability supports the enhancement of diagnostic accuracy, quantitative analysis, and facilitates more reliable and effective decision-making processes in medical applications. Since Ian Goodfellow and colleagues created GANs in June 2014 [1] many variants of GAN flooded the GAN arena.

In this work, the authors have endeavored to comprehensively review various variants of GAN utilized for medical image synthesis/generation, image-to-image translation, denoising & reconstruction, and image segmentation. This research paper explores how GANs are useful in medical image processing and examines recent advancements in GAN technology within this field. The authors assess GAN performance across various datasets encompassing different organs and modalities. Moreover, the study explores prospective advancements in GAN models, aiming to guide future researchers in exploring the potential of GANs in medical image processing.

This paper is structured as follows: Section II provides a concise introduction to Generative Adversarial Networks (GANs). Subsequent subsections (A to D) comprehensively review various GAN variants and pertinent literature concerning their applications in medical image processing, encompassing tasks such as image generation, segmentation, image translation, denoising, and reconstruction. Section III discusses the challenges and implications inherent to GANs. Section IV consolidates the findings and delineates potential avenues for future research.

2 Generative Adversarial Network

GAN is one of the most effective unsupervised techniques in Artificial Intelligence (AI) for synthetic data generation & data augmentation. GAN operates through a dual neural network framework wherein two networks engage in an adversarial process, striving to optimize the fidelity and diversity of generated outputs. These networks are known as generator (convolutional neural network) and the discriminator network (deconvolutional neural network). The generator network is responsible for the fabrication of fake data whereas the discriminator network endeavors to establish a decision boundary that effectively distinguishes between authentic data and synthetic data generated by the generator network. The GAN operates within a zero-sum game framework; when the generator network produces an output of one, it wins and the discriminator network loses, and vice versa. During training feedback loops between the two networks making generator network produce fake data very similar to the ground truth and the discriminator network effectively discriminates the synthetic and generated data. Figure 1 below shows the basic block diagram of GAN.

GANs are used in multiple aspects of medical imaging such as generation, image de-noising and reconstruction, segmentation and image-to-image translation. The delineation of these research domains is expounded extensively in the subsequent sections.

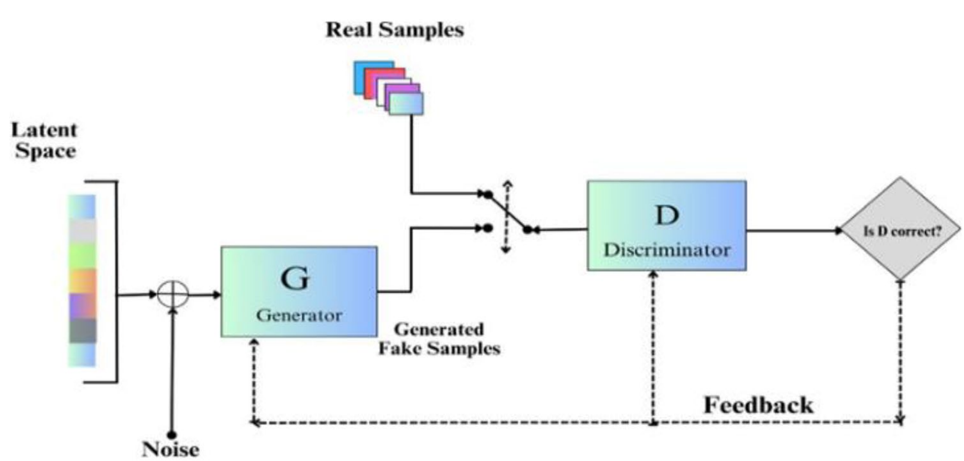

Fig. 1 Shows block architecture diagram of a typical GAN

2.1 GAN in Image Generation

GAN has emerged as a potential tool for medical image generation in the field of medical image analysis. GANs exhibit the capability to generate high-quality medical images, characterized by exceptional structural fidelity and enhanced clinical applicability. Traditional methodologies in medical imaging yield diverse image modalities, including ultrasound, Magnetic Resonance Imaging (MRI), Computed Tomography (CT), and Positron Emission Tomography (PET). The acquisition of these extensive datasets is a labour-intensive and time-consuming endeavour. Consequently, there is an imperative demand for rapid and large-scale data generation. GANs have demonstrated the ability to produce top-notch medical images with superb structural details and increased clinical use. The abundance of medical images improves the accuracy of AI systems for better prediction and diagnosis [2, 3]. Synthetic medical images are generated for tissue recognition on a dataset of 6,000 colonoscopy images using a GAN model [2]. The results showed that the synthetic images generated by the GAN model were able to accurately represent different types of tissues, demonstrating the potential of GANs for medical image synthesis. The model was tested on lung CT images also. Expanding the scope, the potential of GANs was explored for synthetic image generation for other medical imaging domains, including brain MRI, retinal fundus images, and chest X-rays [3]. In the realm of brain MRI [4], focused on generating synthetic brain MR images using GANs. The study focussed on generation of synthetic images that closely resemble real brain MR images, maintaining crucial statistical properties throughout the process.

Moreover, GANs have been leveraged to enhance medical imaging datasets while protecting patient privacy [5] employed GANs to synthesize realistic medical images, enhancing training datasets and ensuring patient information remains anonymous. This study includes classification tests to evaluate how well synthetic images perform when a classifier is trained on a combination of real and synthetic images. Furthermore, GANs have been utilized for the generation of synthetic breast ultrasound images [6]. employed a GAN for synthesis of breast ultrasound images and assessed the quality of the synthetic images using metrics such as the Structural Similarity Index Measure (SSIM) and the Peak Signal-to-Noise Ratio (PSNR). The investigation highlights the ability of GANs to produce high-quality synthetic images in the context of breast ultrasound. The insufficiency of labelled data presents a substantial challenge in medical imaging. With only limited labelled data available, synthetic images are being generated to closely resemble real images [7].

To evaluate the effectiveness of the generated synthetic images, convolutional neural network (CNN) models were employed on liver lesion dataset [8]. It focused on testing how the incorporation of synthetic images affected the performance of the CNN models in image classification tasks. Synthetic Image Generation using GAN often leads to generate a deformed structure and may lose the original shape as of the input image. To overcome such challenges, different approaches were used by introducing new loss functions with GAN [9, 10]. The deformation invariance across medical imaging datasets, including brain MRI and retinal Optical Coherence Tomography (OCT) images is addressed by [9]. Similarly, to preserve the anatomical structures and deformations between the input and generated images of gastric dataset, a GAN based approach with a novel loss function has been employed [10]. Various subjective and quantitative evaluations are employed to assess the quality of the generated images. These evaluations include the Structural Similarity Index, Inception Score, and Fréchet Inception Distance. Figure 2 above and Figs. 3 and 4 below present illustrations from their study showcasing the utilization of various GANs variants alongside the proposed modification for generating synthetic images of Gastritis patches and non-gastritis patches. Images created using the suggested technique are illustrated in Fig. 4 below, which resembles the actual images shown in Fig. 2 above. Controllability over the generation process is essential for creating medical images since it enables researchers to alter certain aspects of the images, such as the existence of tumours, the size of organs, or the level of contrast. This issue was addressed by [11]. To enable the modification of particular features of generated images, an encoder is added to encode certain image attributes in latent code which is finally provided as input to the GAN generator. The system enables manipulation of the generated image by altering the latent code via the encoder. Table 1 below contrasts the techniques and findings of different research studies.

2.2 GAN in Segmentation

Medical image segmentation is an important task in medical imaging that involves distinguishing different structures or regions of interest within medical images. Precise segmentation of medical images holds paramount importance in various clinical applications, including diagnosis, treatment planning, and disease progression monitoring. In segmentation tasks, the generator of a GAN is trained to receive an input image and produce a segmentation mask that assigns a class label to each pixel in the image. One of the advantages of using GANs for segmentation is its potential to generate high-resolution output which is visually appealing with fine details [12]. segmented skin lesions in dermoscopy images

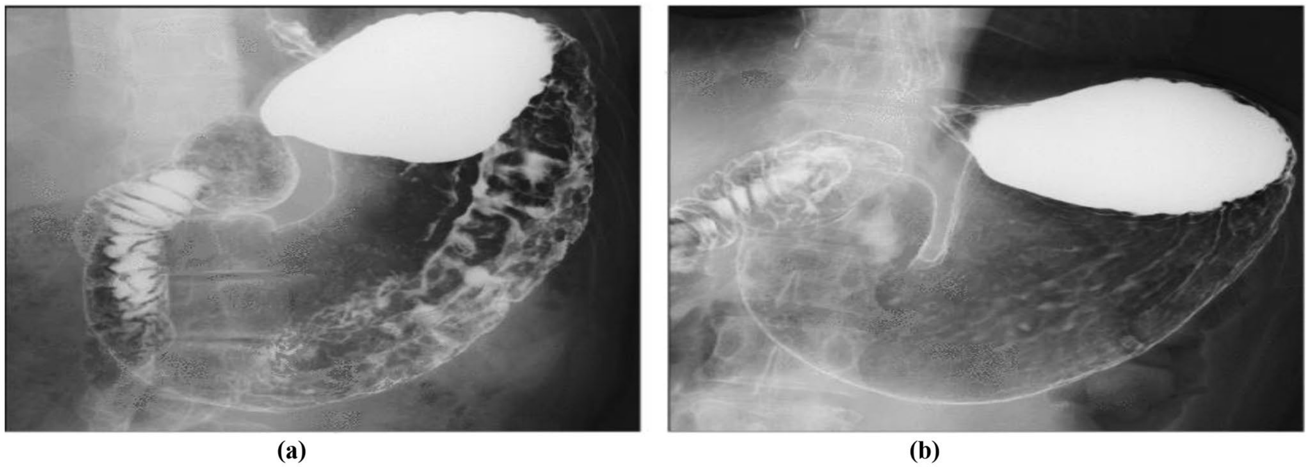

Fig. 2 Adapted from [2]: shows gastric X-ray images used in their study: (**a**) an image with gastritis and (**b**) an image without gastritis (non-gastritis)

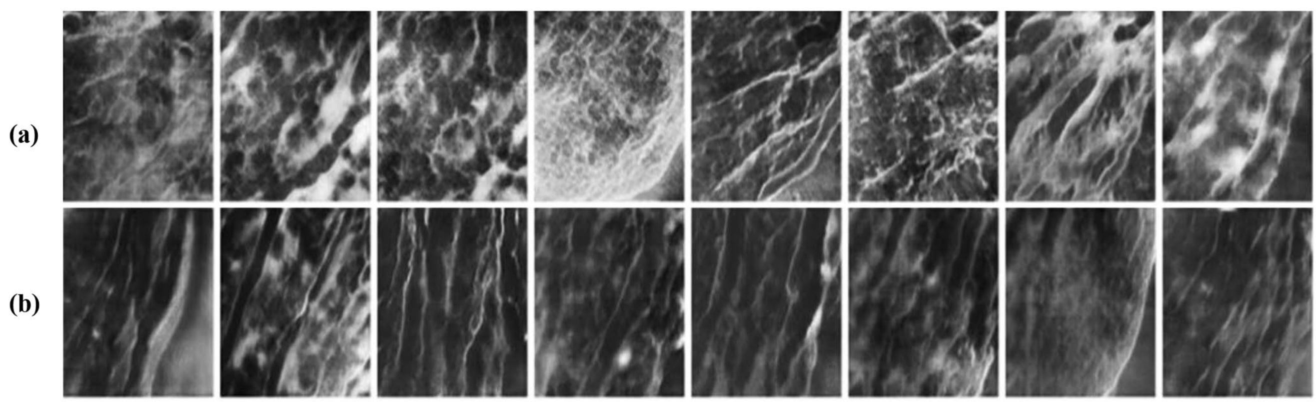

Fig. 3 Adapted from [10]: shows extracted patches of Gastritis (299 × 299 pixels). (**a**) Generated gastritis patches (**b**) Generated non-gastritis patches

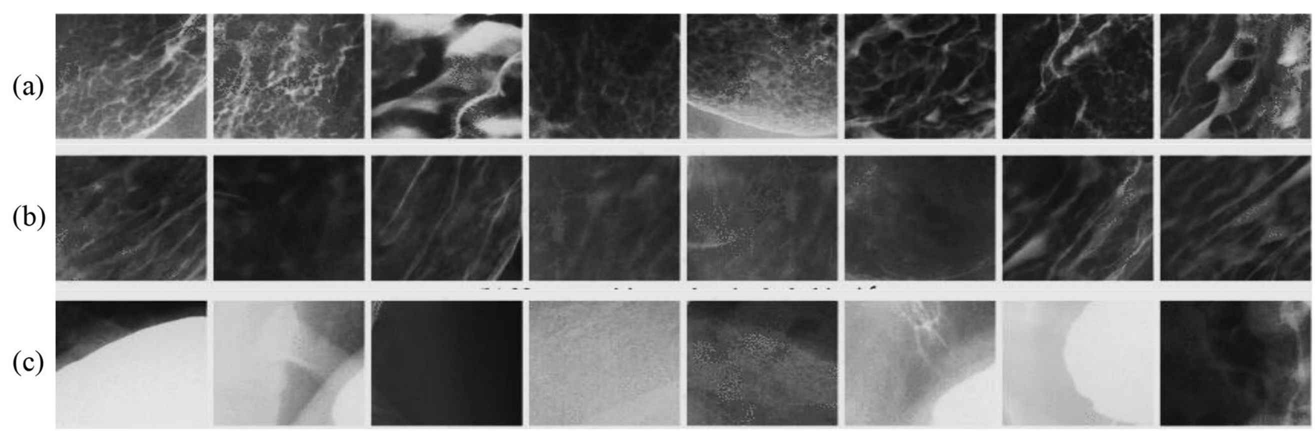

Fig. 4 Adapted from [10]: shows patches generated by their model LC-PGG. (**a**) Gastritis patches included in A (**b**) non-gastritis patches included in N. (**c**) Outside patches included in O

using the GAN-based approach by training a GAN model with dual discriminators, one for the segmentation mask and the other for the resulting picture. GANs when fused with CNN segmentation networks [13–15], produced notable results for two publicly available datasets for skin lesions, the International Skin Imaging Collaboration (ISIC) 2018 and the dermoscopic image database acquired from the dermatology service of hospital Pedro Hispano (PH2).

Table 1 Different studies conducted in medical image "Generation/Synthesis" using different variants of GAN

S. No.	Reference and Year	Medical Image used	GAN Architecture	Loss function	Main Contribution
1.	[4] 2018	Brain MRI	3D GAN	L1 loss and adversarial Loss	Synthetic 3D brain MR images were generated, that can be used for data augmentation and image analysis tasks.
2.	[7] 2022	Multiple modalities	Deep Pixel to Pixel GAN (Pix2Pix GAN)	L1 loss and adversarial loss	Modified Pix2Pix GAN architecture for generating high-quality synthetic medical images that can be used for data augmentation.
3.	[9] 2021	Multiple modalities	Deformation Invariant CycleGAN (DiCyc)	Adversarial loss, cycle consistency loss	Deformation invariant cross-domain information fusion for medical image synthesis.
4.	[2] 2018	Histo-pathol-ogy images	Conditional GAN(CGAN)	Binarycross-entropy loss	GAN-based method for synthesizing medical images to improve tissue recognition.
5.	[3] 2018	Multiple modalities	Multi-Generator Adversarial Network. (MI-GAN)	Adversarial loss, mean squared error loss	Demonstrating the ability of GAN to generate high-quality and diverse medical images that can be useful for various applications in healthcare and medical research.
6.	[8] 2018	Liver CT scans	Modified Deep Conditional GAN (DCGAN)	Adversarial loss, mean squared error loss	Demonstrating the effectiveness of GAN-based synthetic medical image augmentation for improving the performance of CNNs in liver lesion classification.
7.	[5] 2018	Magnetic Resonance Imaging (MRI)	DCGAN	Wasserstein GAN loss with gradient penalty	Presented a method for synthesizing realistic medical images to increase the size of the training dataset using a GAN to generate brain MR images.
8.	[6] 2019	Breast ultrasound images	DCGAN	Binarycross-entropy loss	Synthetic breast ultrasound images were generated to increase the size of medical image datasets and improve machine learning algorithm performance
9.	[10] 2019	Gastritis patches	New Loss function-based Progressive Generative Adversarial Network (PGGAN)	Novel loss function (based on both Mean Squared Error (MSE) and adversarial loss. that emphasizes the semantic content and structure of the image	Generated diverse and realistic synthetic gastritis images that are useful for medical education and research.
10.	[11] 2022	Brain MRI	GANbased novel approach for spatial transformation	Feature matching loss	Introduced a controllability module to generate medical images with specific features, such as lesion location and size, to facilitate medical diagnosis and treatment planning

In [13] U-Net architecture combined with GAN-based discriminator is employed to automatically segment multiple organs in thoracic CT images. A sizable dataset of thoracic CT scans with manually labelled organ segmentations served as the basis for training the U-Net-GAN model. Figures 5 and 6 below are the illustrations from the suggested. U-Net architecture is combined with a GAN and employed a multi-scale L1 loss function named as Segmentation and Adversarial Network (SegAN) [14].

By utilizing the benefits of both the U-Net architecture and the GAN, a novel loss function known as the Adversarial Confidence Loss has been utilized by [15]. Another approach in GAN-based Segmentation is incorporation of pre-processing and transfer learning with GAN for better efficiency which segmented two publicly accessible datasets of retinal blood vessels [16]. The images were pre-processed using Automatic Colour Equalization (ACE) to enhance the retinal blood vessels more clearly. To further boost training effectiveness and strengthen segmentation robustness, binary cross-entropy loss function and the false-negative loss function were used. Incorporating transfer learning with GAN proved effective in segmenting unlabelled datasets [17]. The study applied a pre-trained edge detector based on Richer Convolutional Features (RCF) and Visual Geometry Group 16 (VGG), to create straightforward edge diagrams from genuine unlabelled training images. A GAN is then trained to create artificial medical images from the edge diagrams using the associated image-diagram pairs. Further, a method was introduced for cell image segmentation using GANs and transfer learning [18]. To address the limited availability of annotated data for cell segmentation, the proposed strategy integrates the capabilities of GANs in generating synthetic data alongside transfer learning techniques. Another approach in GAN-based segmentation is to combine both supervised and unsupervised learning strategies, making a multi-stage setup [19]. Deep learning methods have shown great promise in medical image segmentation; however, these are still limited by the availability and quality of annotated data. Therefore, there is always a need for human intervention for refinement [20]. A hybrid

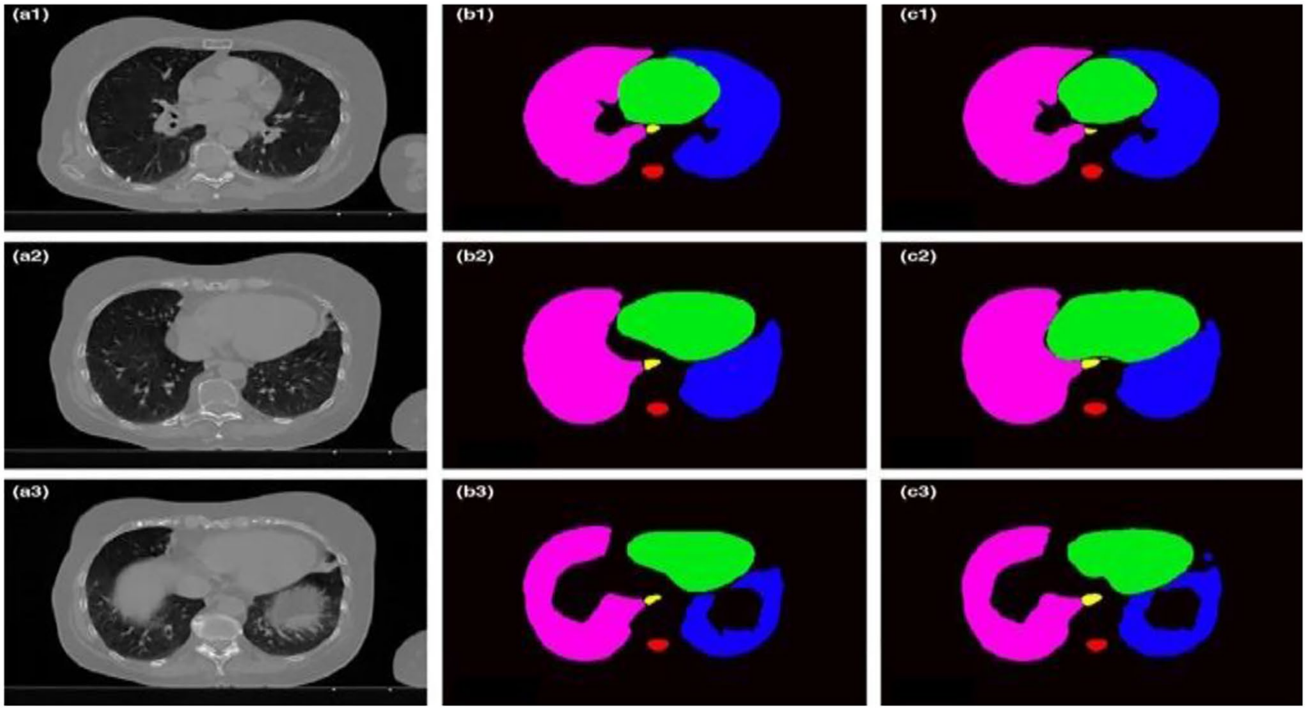

Fig. 5 Adapted from [13]: (**a**) shows three transverse CT slices of a single patient together with the organ-at-risk contours that were determined using (**b**) manual contouring (ground truth) and (**c**) the suggested method

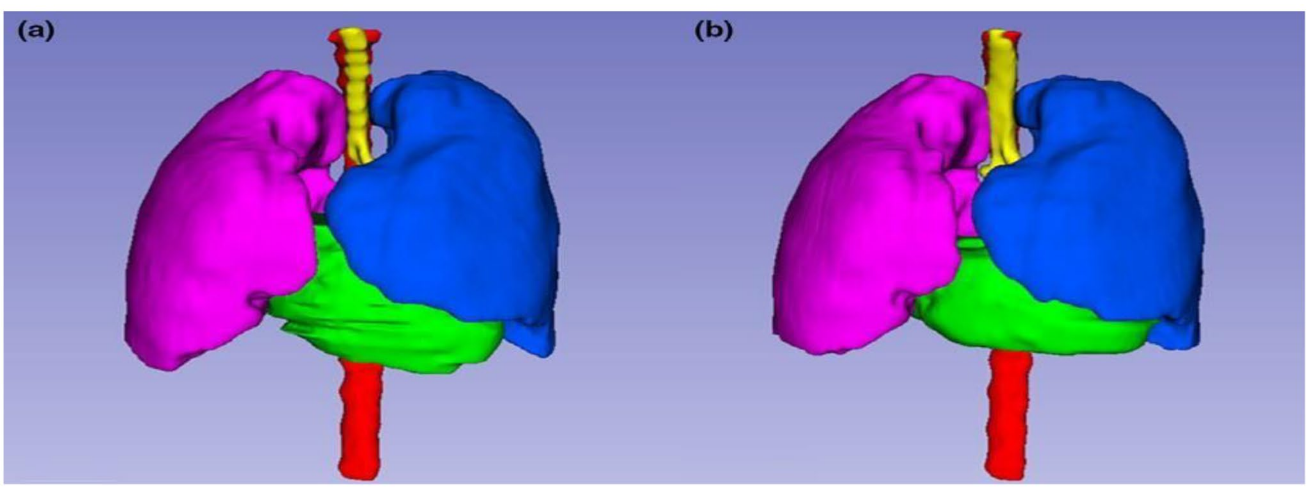

Fig. 6 Adapted from [13]: A 3D visualization of the organ-at-risk outlines on the same patient in Fig. 5 obtained with (**a**) manual contouring and (**b**) the suggested method

method integrating machine intelligence with human expertise enhances machine-generated segmentation masks to approximate the corrections performed by human experts. GANs are also used in 3D image segmentation [21] segmented publicly available 3D brain datasets using a method for few-shot 3D multi-modal medical image segmentation with generative adversarial learning.

With a sparse set of annotated examples, the proposed methodology integrates a segmentation network with a generative adversarial network for the segmentation of medical images. The techniques and findings of the various researches are contrasted in Table 2 below.

2.3 GAN in Image-to-Image Translation

Clinical diagnosis and treatment hinge significantly upon medical imaging, with the ability to generate high-quality images across different contrasts playing a vital role in

Table 2 Studies conducted in medical image "Segmentation" using different variants of GAN

S.no	Reference and Year	Type of Medical Image	GAN Architecture	Loss function	Main Contribution
1.	[12] 2020	Skin lesions	GANwith dual discriminators	Adversarial loss, perceptual loss, and segmentation loss	Introduced a novel GAN architecture with dual discriminators and a multi-scale generator for skin lesion segmentation.
2.	[13] 2019	Thorax CT images	U-Net + GAN	Adversarial loss, Dice loss, andGradient penalty	Automatic multiorgan segmentation method for thorax CT images using a U-Net-GAN.
3.	[16] 2020	Retinal images	Stackeddeep fully convolutional networks with a GAN (MGAN)	Combination of Dice loss and binary cross-entropy loss	Introduced a novel loss balancing method and a stacked deep fully convolutional network with a GAN for accurate segmentation of retinal blood vessels.
4.	[18] 2019	Cell Images	U-Net + GAN	Binary Cross-Entropy Loss	Efficient and effective segmentation of cell images using U-Net and GANs.
5.	[19] 2022	Brain MRI images	Multi-stage GAN	Binarycross-entropy and adversarial loss	Improved segmentation accuracy and efficiency using multi-stage GAN architecture and multiple specialized models.
6.	[21] 2018	Brain tumour images	Segmentation network with GAN	Binary cross-entropy and adversarial loss with dice loss	Improved few-shot segmentation accuracy for multi-modal medical images using generative adversarial learning and a novel data augmentation method.
7.	[15] 2020	Retinal images and brain tumour images	Segmentation network with GAN	Adversarial confidence loss and cross-entropy loss	Introduced a novel adversarial confidence-learning framework for medical image segmentation and synthesis that explicitly models the model's level of confidence in its predictions, improving the robustness and accuracy of the segmentation findings.
8.	[20] 2020	Retinal and Cardiac Images	DCGAN	Collaborative Loss	Developed a hybrid approach combining machine learning and human experts to improve medical image segmentation accuracy and efficiency.
9.	[14] 2018	Brain MRI images	SegAN (GAN + U-Net) Segmentation and Adversarial Network	Multi-scale L1 loss	Introduced a novel Segmentation framework that combines a U-Net architecture with a GAN and employs a multi-scale L1 loss function for medical image segmentation.
10	[17] 2019	Multiple modalities	GAN based	Adversarial loss, Edge loss, Total variation loss	Developed an adversarial network-based unsupervised method for segmenting medical images that accepts edge diagrams as input and outputs segmentation maps.

accurately identifying and treating patients. Traditional picture translation techniques are labour-intensive, time-consuming, and costly since these rely on handmade characteristics. Diverse image standards employed in medical imaging produce different outcomes. In certain scenarios, these standards provide complementary information, thus requiring simultaneous reference to ensure precise diagnostic accuracy. In the field of medical science, hybrid imaging is currently used for such purposes. GANs can be used to create new images or convert existing ones from one input standard to another. GANs' are trained to convert an image from one domain (such as an MRI scan) to another domain in (such as a CT scan) [22]. The translation of images between modalities that might not often be compatible is another possible application of GANs in medical imaging. The work of [23] on brain images, where the PET standard images were translated to CT images, is one significant example. GAN-based image-to-image translation technique when fused with a deep learning algorithm produces effective results as shown in Fig. 7. Furthermore, low-quality endoscopic images [18] were introduced into other modalities by employing a multiscale discriminative network and a U-Net generator with residual blocks. Image-to-image GAN when combined with noise-to-image GAN also helped in synthesis process [25]. Their method involved training a GAN to create synthetic brain MR images that closely match actual images using noise patterns generated by a trained noise GAN. Different loss functions, such as an adversarial loss, perceptual loss, and mean squared error loss were used to produce realistic images. The GAN-based image translation proved an effective technique to augment the dataset of medical conditions, which are in scarcity at the current time. For example, the problem of insufficient COVID-19 training data, which limited the performance of traditional machine learning models [26, 27]. Moreover, a publicly available dataset of COVID-19 chest X-rays was used [26] to assess the efficacy of a trained model to convert the input chest X-ray images into their corresponding

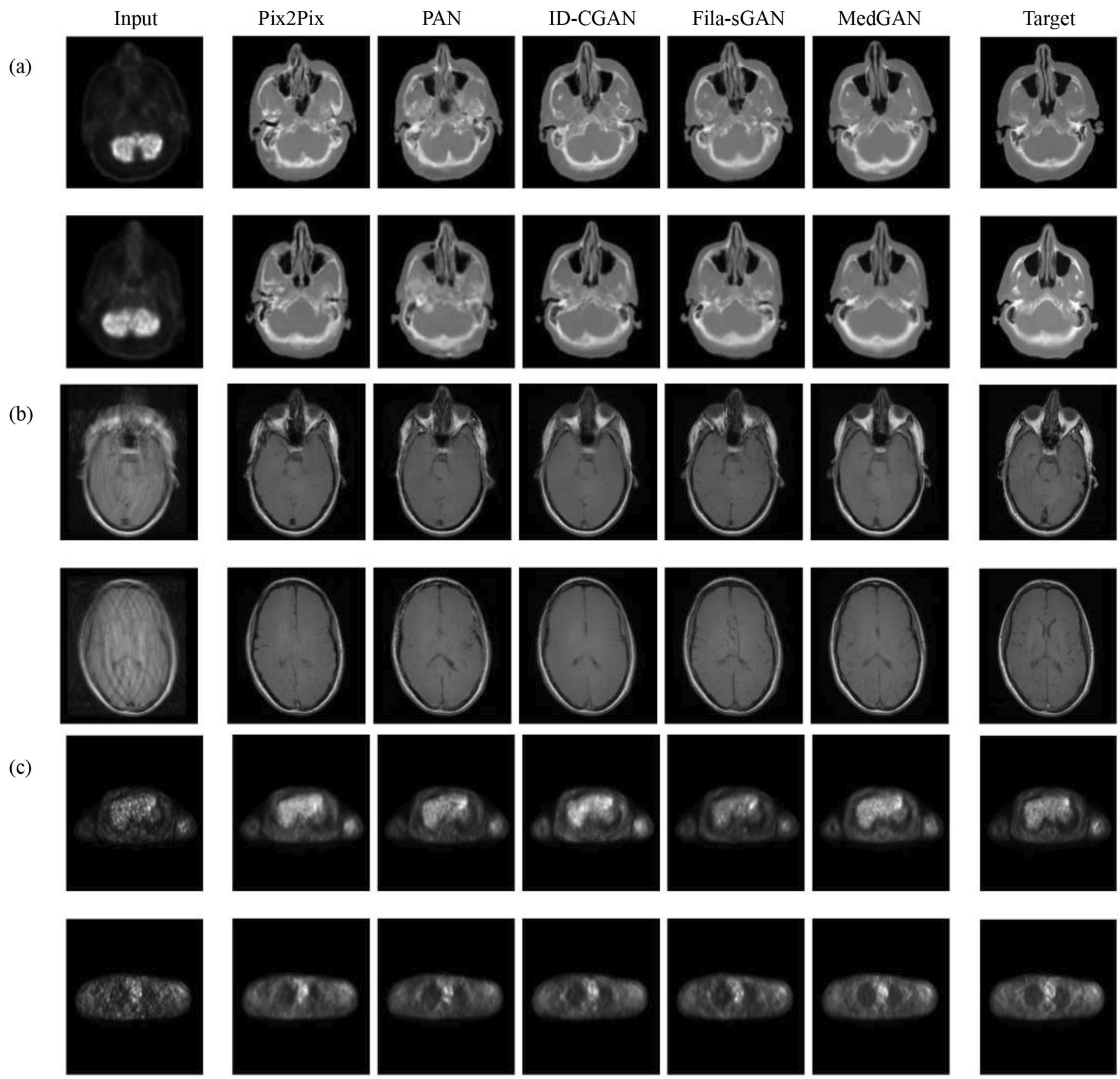

Fig. 7 Adapted from [23]: shows the proposed method result in image augmentation. (**a**). PET-CT translation (**b**). MR motion correction (**c**) PET denoising

COVID-19 positive or negative versions. Further, another approach was introduced [27] for synthesizing COVID-19 chest X-rays from normal chest X-rays using unpaired image-to-image translation. The presented technique employed a cycle consistency loss to make sure that the mapping is consistent in both directions and has two generators and two discriminators, one for each domain. Maintaining the integrity of image contrast features post-translation across different modalities is crucial in medical imaging and represents a formidable challenge in the domain of image-to-image translation. GANs proved to be effective in such cases. By adding a second contrast classification network to the GAN design, the issue of MR image translation across various contrast modalities is solved [28, 29]. Further, MR-contrast-aware image-to-image translation was performed [28] using GANs and evaluating it on a dataset of brain MR images. Moreover, a comparison of two deep learning models was performed [29] for image-to-image translation of multi-contrast (MR) images. The two models, Cycle-Consistent Generative Adversarial Networks (CycleGAN) and Unsupervised Image-to-Image Translation (UNIT) utilized for image-to-image translation were both capable of translating multi-contrast MR images from one image to another in their findings. In image translation, the input images may contain additional details besides the region of interest. Therefore, eliminating or suppressing these details in the output during image-to-image translation is another GANs' application area. GANs produced impressive results in the suppression of unwanted details other than the region of interest. GAN employed in chest X-ray radiography [30] eliminated the bone structure from X-ray images and successfully performed the segmentation process of the lungs.

Unpaired image translation entails training a model with two separate sets of images, one from each domain, without any one-to-one correspondence in between. Establishing significant correspondences between images originating from disparate domains without explicit image pairs presents a formidable challenge. Consequently, this predicament often leads to the model producing translations that are either unrealistic or inconsistent. This issue is addressed by [31] creating surgical training phantoms that are more realistic by using deep unpaired image-to-image translation. The pairs of real surgical images were used and matching synthetic images were produced using a conventional 3D printing technique to train the GAN. The techniques and findings of the various studies are compared in Table 3 below.

2.4 GAN in Image Denoising/Reconstruction

Reconstructing and denoising medical images are critical tasks within the imaging domain. The accuracy of medical imaging directly influences patient outcomes, treatment planning, and diagnostic precision. In recent years, a substantial surge in the deployment of deep learning techniques for the purposes of medical image reconstruction and denoising is evident. Traditional medical imaging procedures have the inherent drawback of producing pictures with noise. Also, medical images with low resolution and noise can affect the precision and efficacy of medical image analysis and diagnosis. However, Generative Adversarial Networks (GANs) emerge as a promising alternative for mitigating noise in medical imaging. By converting low-resolution medical images into high-resolution ones, the GAN technique increases the amount of detail and quality. In one of the works, GAN was used for denoising low-dose chest CT images using CGAN [32]. The encouraging improvements have been done in image quality and noise

Table 3 Different studies conducted in medical image "Image-to-image Translation" using various variants of GAN

S.no	Paper and Year	Type of Medical Image	GAN Architecture	Loss function	Main Contribution
1.	[23] 2020	X-ray, CT, and MRI images	DCGAN and CycleGAN	Adversarial and cycle consistency Loss	Developed a GAN-based model that can perform multi-modal medical image translation.
2.	[28] 2021	MRI	Contrast-aware Cycle GAN	L1 L2, Adversarial loss, perceptual loss	Presented a GAN model that can effectively perform contrast-aware image-to-image translation in MRIs, improving image quality and maintaining clinically relevant features
3.	[24] 2020	Endoscopic images	Live-Cadaver Generative Adversarial Network (LC-GAN)	L1, L2, SSIM, perceptual loss	Enables image-to-image translation of endoscopic images for various modalities and devices.
4.	[22] 2020	MRI images	CycleGAN	L1, SSIM	Translates MRIs from one modality to another, while preserving image content and enhancing structural details.
5.	[29] 2018	BrainMR images	CycleGAN,+UNIT	Adversarial loss, Cycle consistency loss, Content loss	Investigated the use of GANs on multi-contrast MR images for image-to-image translation and compared the performance of two GAN models.
6.	[25] 2019	BrainMR images	Combination of noise-to-image GAN and image-to-image GAN	Adversarial loss, perceptual loss, mean squared error loss	Developed a novel approach for augmenting brain MR images to improve the detection of brain tumours.
7.	[30] 2020	ChestX-ray radiography	Pix2pix GAN	Adversarial loss, L1 loss, Perceptual loss, dice loss	Introduced a method for simultaneously performing bone suppression and organ segmentation in chest X-ray radiography using an image-to-image translation approach with multiple loss functions.
8.	[27] 2021	ChestX rays (normal and COVID-19)	Dual Generative Adversarial Networks (DualGAN)	Cycle consistency loss	Synthesis of COVID-19 chest X-rays from normal chest X-rays using unpaired image-to-image translation with attention mechanism.
9.	[26] 2020	Chest X-rays	CGAN	Adversarial loss, Cycle consistency loss	Enhancing the performance of automated COVID-19 chest X-ray diagnosis by using image-to-image translation to synthesize COVID-19 chest X-rays from normal chest X-rays.
10.	[31] 2019	Surgical images	CycleGAN	Adversarial loss, cycle consistency loss, identity loss	Developing a method for generating hyperrealistic surgical images using deep unpaired image-to-image translation.

reduction. GAN trained on paired noisy and denoised single-photon emission computed tomography (SPECT) pictures, also produced notable results for denoising SPECT images acquired using low radiation doses [33]. The network was able to create high-quality denoised images from low-dosage input images by learning a mapping between the two types of images. A combination of Wasserstein distance and perceptual loss with GANs has also improved the denoising capability for low-dose CT images [34]. The Wasserstein distance ensured that the generated images are statistically similar to the real ones. Employing adversarial loss function for training of GAN and perceptual loss, to quantify the structural and semantic differences between high-resolution pictures at the feature level, produced high-resolution images with better clarity, sharpness, and finer details [35]. The traditional imaging techniques introduce poison noise in medical X-ray images which impairs the diagnostic efficacy and thus hinders subsequent analysis. The GAN-based approach with the Wiener filter [36] adaptively suppresses the noise while keeping account of both the statistical characteristics of the noise and the signal-to-noise ratio of the input image data. Using a GAN framework with other learning methods resulted in medical images with enhanced resolutions which gave more in-depth data for analysis and diagnosis in the medical field. Further in [37], a variety of learning techniques, including identical learning, residual learning, and cycle learning were combined with GAN for CT image super-resolution. The cycle learning restriction preserved the important structural information by imposing a residual learning constraint to focus on features absent from the low-resolution picture. Fusion of different loss functions like adversarial loss and pixel-wise L1 loss was also utilized to train the model. The pixel-wise L1 loss promoted pixel-level similarity, whereas the adversarial loss encouraged the generator to produce high-resolution images [38] introducing a novel approach for medical image super-resolution using a GAN-based method. Figure 8 below shows the output of the method introduced. PET imaging finds widespread application in medical diagnostics, necessitating the administration of a radiotracer for adequate image resolution. Nonetheless, the potential for elevated radiation exposure poses a significant apprehension among patients, thereby constraining the practicality and frequency of PET scans. To address this issue, a solution based on GAN was presented that aimed to reconstruct PET images from ultra-low-dose data, reducing the radiation dose required for imaging [39]. The framework also incorporated feature matching and task-specific perceptual loss into the training process. The class imbalance problem is the greatest hindrance to a reliable diagnostic system. Additionally, a GAN-based framework for medical image reconstruction was introduced to address the class imbalance problem in Alzheimer's disease assessment [40]. The presented technique transformed the low-quality input images into high-quality brain MRI scans using a GAN. Likewise, a GAN-based method was presented for medical image reconstruction for the publicly available Brain Tumour Image Segmentation (BRATS) dataset [41]. Deep image reconstruction from unregistered measurements without relying on ground truth data is also possible as mentioned in [42, 43]. It addressed this issue by putting forth a deep learning-based method that without the need for ground truth images, can directly recreate high- quality images from unregistered data. The presented technique learns the mapping between the unregistered measurements and the relevant picture using a GAN. Consequently, the network generates precise reconstructions and learns the underlying picture properties without explicit reliance on registered measurements or ground truth data. The techniques and findings of the various researches are compared in Table 4 below.

3 Challenges and Discussion

The application of GANs in the generation of medical images is currently gaining considerable attention within the domain of medical image analysis. GANs are proving to be a promising tool for producing high-fidelity medical images that exhibit exceptional structural integrity, thereby rendering them applicable across a spectrum of clinical scenarios. Traditional methods of obtaining medical images from different modalities involve time-consuming and challenging processes, often resulting in limited datasets. However, GANs offer a solution by generating synthetic medical images in large quantities and short periods [44]. The application of GANs in medical image generation encompasses different imaging modalities such as ultrasound, MRI, CT, X-ray, PET, and more. The generated images are applicable for a myriad of tasks encompassing disease diagnostics, image denoising, image augmentation, and tackling the issue of scarce labelled data in healthcare applications.

Numerous studies have substantiated the efficacy of GAN-based methodologies in the generation of medical images. Researchers have adeptly produced synthetic medical images that accurately depict various tissue types, encompassing colonoscopy images and lung CT images. Additionally, the capability to generate synthetic brain MR images has been demonstrated to faithfully preserve the original anatomical structures and deformations. Certain challenges related to GAN based image processing have been identified as: (i) Some studies have highlighted the sensitivity of GANs to noisy or poor-quality medical images, resulting in unreliable results during training. This challenge poses difficulties in achieving consistent and

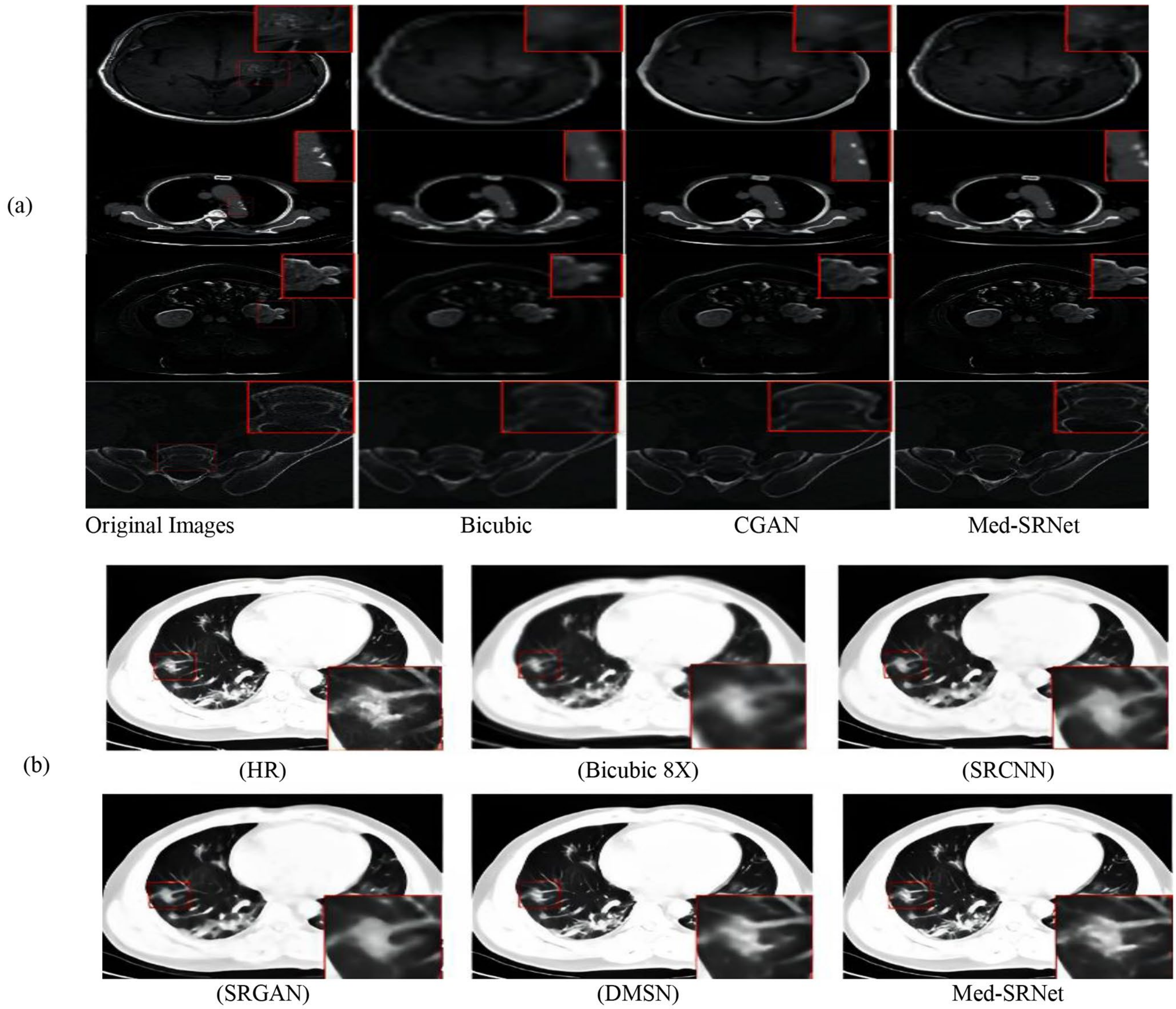

Fig. 8 Adapted from [38]: showing the Qualitative result of their study when used on (**a**) brain images and (**b**) lung images

accurate outcomes. (ii) Training GANs require extensive and diverse datasets that represent various medical imaging scenarios. However, such datasets are often limited, hindering the training process and reducing the generalizability of GAN models. (iii) Traditional evaluation metrics such as Mean Square Error (MSE), PSNR, and SSIM do not fully capture the visual quality perceived by human eyes. These metrics fail to account for the visual properties and nuances of medical images, necessitating the development of more robust evaluation metrics. (iv) During cross-domain image-to-image translation, where unpaired training data is utilized, there is no guarantee that small anomaly regions will be preserved. The absence of a data fidelity loss term in unpaired training can lead to the loss of crucial details during image translation.

GANs are presently employed in medical image segmentation, a critical component in numerous clinical applications. This integration results in precise segmentation of diverse structures or regions of interest within medical images. GANs have the advantage of generating high-resolution output with fine details, improving the precision and effectiveness of segmentation tasks [45]. These segmentation tasks have been applied to various medical imaging domains, including skin lesion segmentation, organ segmentation in thoracic CT images, and retinal blood vessel segmentation, among others. Moreover, GANs are utilized for image-to-image translation within the domain of medical imaging. This application encompasses the transformation of images across diverse modalities, exemplified by converting MRI scans into CT scans, improving the clarity

Table 4 Different studies conducted in medical image "Denoising/Reconstruction "using various variants of GAN

S.no	Paper and Year	Type of Medical Image	GAN Architecture	Loss function	Main Contribution
1.	[32] 2020	ChestCT Images	CGAN	Adversarial loss, L1 loss	Denoising low-dose CT images
2.	[33] 2022	Myocardial Perfusion SPECT Images	Pix2Pix GAN	Adversarial loss, L1 loss	Denoising low-dose myocardial perfusion SPECT images
3.	[40] 2020	BrainMRI Images	DCGAN	Binary cross-entropy loss	Addressing class imbalance problem in Alzheimer's disease diagnosis using GAN-based image reconstruction
4.	[41] 2020	BrainMRI Images	Medical Image Reconstruction Generative Adversarial Network. (MIRGAN)	Adversarial loss, Perceptual loss,	Medical image reconstruction using multi-loss GAN (MirGAN) for improved image quality
5.	[34] 2018	CT images	Wasserstein Generative Adversarial Network. (WGAN)	Wasserstein distance and perceptual loss	Developed a novel approach for low-dose CT image denoising using a conditional Wasserstein GAN architecture with Wasserstein distance and perceptual loss.
6.	[35] 2022	Multiple modalities	Super-resolution Generative Adversarial Network. (SRGAN)	Perceptualloss, content-loss, and adversarial loss	Introduced a new GAN-based approach for medical image super-resolution that outperforms existing methods.
7.	[36] 2022	ChestX-ray images	CGAN.	Adversarial loss	A novel Wiener filter-based method for denoising Poisson noise from medical X-ray images.
8.	[38] 2022	Multiple modalities	GAN-based Medical Super Resolution Network (SRNet.)	Adversarial Loss, L1 Loss	GAN-based method for medical image super-resolution using high-resolution representation learning.
9.	[39] 2019	Ultra-low-dose PET	CGAN	Task-specific perceptual loss.	Reconstruction of high-quality PET images from ultra-low-dose data, reducing radiation exposure during PET scans.
10.	[37] 2019	CT images	residual CNN-based networking with the CycleGAN framework	Adversarial loss, cycle-consistency loss, identity loss, and joint sparsifying transform loss.	Proposes a CT super-resolution method using a GAN framework with multiple learning constraints

of endoscopic imagery, and generating COVID-19 chest X-rays from standard chest X-ray images. Additionally, GANs find application in translating MR image contrasts and fabricating surgical training models that offer lifelike visual representations for educational purposes in surgery.

The integration of GANs in medical imaging research can enhance the accuracy of AI systems, improve patient care, and facilitate various clinical applications that heavily rely on medical imaging [46, 47]. Continued advancements and research in this field have the potential to revolutionize medical imaging practices and contribute to better healthcare outcomes. (i) To bridge the gap between research and clinical practice, further studies are required to establish a practical background for GANs in medical imaging. Rigorous investigations and collaborations between researchers and medical professionals are crucial to address the specific needs and challenges of clinical implementation. (ii) Collaboration and dialogue between researchers and medical professionals are essential to ensure the successful integration of GANs in medical imaging. By understanding the practical requirements and limitations, GAN-based solutions can be developed that are both applicable and relevant in clinical settings. (iii) To overcome challenges related to input data sensitivity and limited datasets, the development of enhanced GAN training methods is necessary. Techniques such as transfer learning, domain adaptation, and data augmentation should be explored to improve the robustness and generalizability of GAN models. (iv) There is a need to develop more reliable GAN models that can handle cluttered and subpar medical images, including those with noise and artifacts. This requires the improvement of GAN architectures and training strategies to ensure better performance and applicability in medical imaging tasks.

4 Conclusion

The application of Generative Adversarial Networks (GANs) in medical image generation is revolutionizing medical image analysis by producing high-quality synthetic images across various modalities, including ultrasound, MRI, CT, X-ray, and PET. Unlike traditional methods that are time-consuming and often yield limited datasets, GANs can swiftly generate vast quantities of detailed medical images, aiding in diagnostics, image denoising, augmentation, and addressing the challenge of limited labeled data. Their efficacy extends to accurate segmentation tasks and image-to-image translation, enhancing applications such as skin lesion and organ segmentation, and modality conversion. This advancement promises to significantly improve AI systems' accuracy in medical imaging, patient care, and various clinical applications. However, practical implementation in clinical settings demands further research and collaboration between researchers and medical professionals to refine GAN models, address data sensitivity issues, and enhance model robustness and generalizability. Continued efforts in this field are essential for stabilizing GAN models and ensuring their effective integration into clinical practice, ultimately advancing disease detection and treatment.

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